

COMPUTER NETWORK Model Pratical

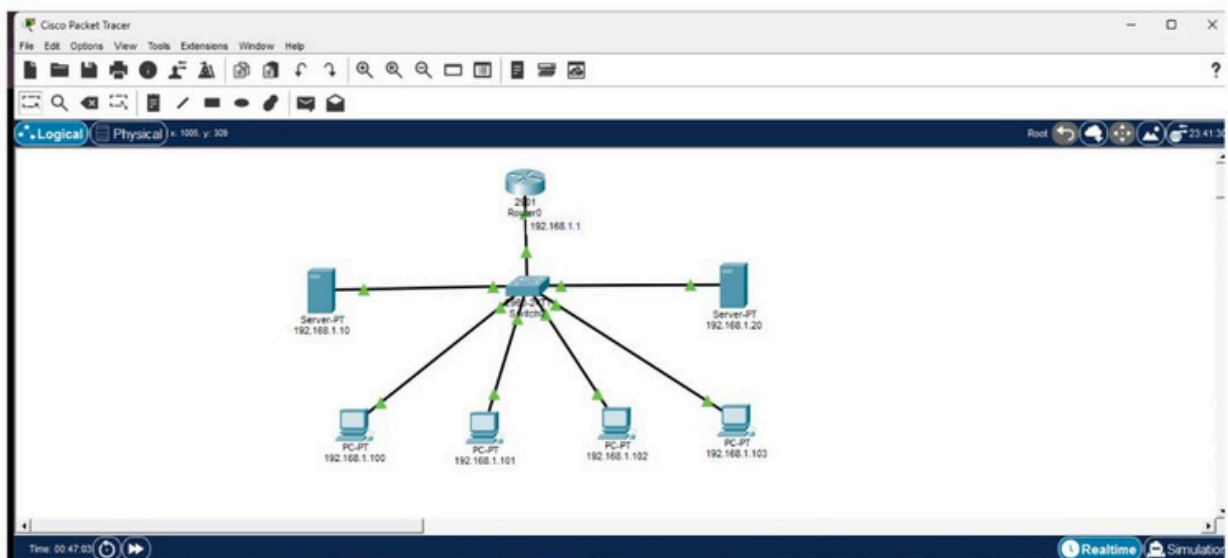
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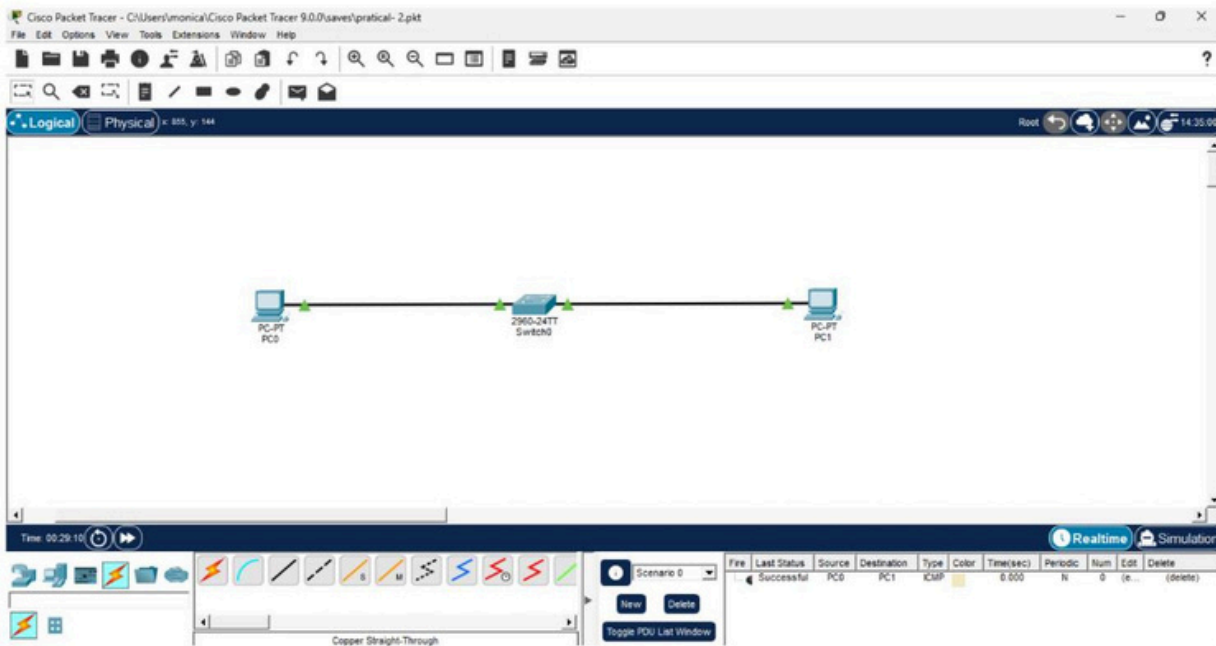
SET 4

1. Assume that you are the network administrator of a data center requiring load balancing across 2 servers. Design and configure the network using Packet Tracer to demonstrate basic load distribution.
2. Configure Port Security on a switch in Packet Tracer to allow only specific MAC addresses to connect to a port, and verify that unauthorized devices are blocked.
3. Using Wireshark, Capture and analyze the packets generated during a Traceroute operation (ICMP TTL Exceeded) for different header fields and payload of the concerned packet.
4. Using C program, Implement and demonstrate parity bit generation and checking for error detection on a given data.

MODEL PRACTICAL EXAM QUESTION 1:



QUESTION - 2



QUESTION 3:

Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
315	9.731521500	172.25.102.49	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=252/64512, ttl=1 (no response found!)
328	9.896158600	172.25.96.1	172.25.102.49	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
329	9.898400300	172.25.102.49	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=253/64768, ttl=1 (no response found!)
330	9.901978400	172.25.96.1	172.25.102.49	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
331	9.903466400	172.25.102.49	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=254/65024, ttl=1 (no response found!)
332	10.071580000	172.25.96.1	172.25.102.49	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
363	10.749054200	172.25.96.1	172.25.102.49	ICMP	70	Destination unreachable (Port unreachable)
417	12.081862000	172.25.96.1	172.25.102.49	ICMP	70	Destination unreachable (Port unreachable)
520	15.919360600	172.25.102.49	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=255/65280, ttl=2 (no response found!)
521	15.934461600	172.22.10.1	172.25.102.49	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
522	15.936078600	172.25.102.49	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=256/1, ttl=2 (no response found!)
523	16.148750600	172.22.10.1	172.25.102.49	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
524	16.150061800	172.25.102.49	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=257/257, ttl=2 (no response found!)
525	16.169725000	172.22.10.1	172.25.102.49	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
529	16.211782800	172.22.10.1	172.25.102.49	ICMP	70	Destination unreachable (Port unreachable)

> Frame 315: Packet, 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface V

> Ethernet II, Src: MNC_08:4d:29 (0c:0b:5b:80:4d:29), Dst: Cisco_6d:c0:e7 (24:d5:e4:6d:c0:e7)

> Internet Protocol Version 4, Src: 172.25.102.49, Dst: 8.8.8.8

> Internet Control Message Protocol

0000 24 d5 e4 6d c0 e7 8c 0b 5b 88 4d 29 08 00 45 00 \$..m....[.M)..E-

0010 00 5c 21 f0 00 00 01 01 75 4c ac 19 66 31 08 00 .\!.....uL..f1..

0020 00 00 00 00 f7 02 00 01 00 fc 00 00 00 00 00

0030 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0040 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0050 00 00 00 00 00 00 00 00 00 00 00 00 00 00

0060 00 00 00 00 00 00 00 00 00 00 00 00 00 00

```

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\monica>tracert 8.8.8.8

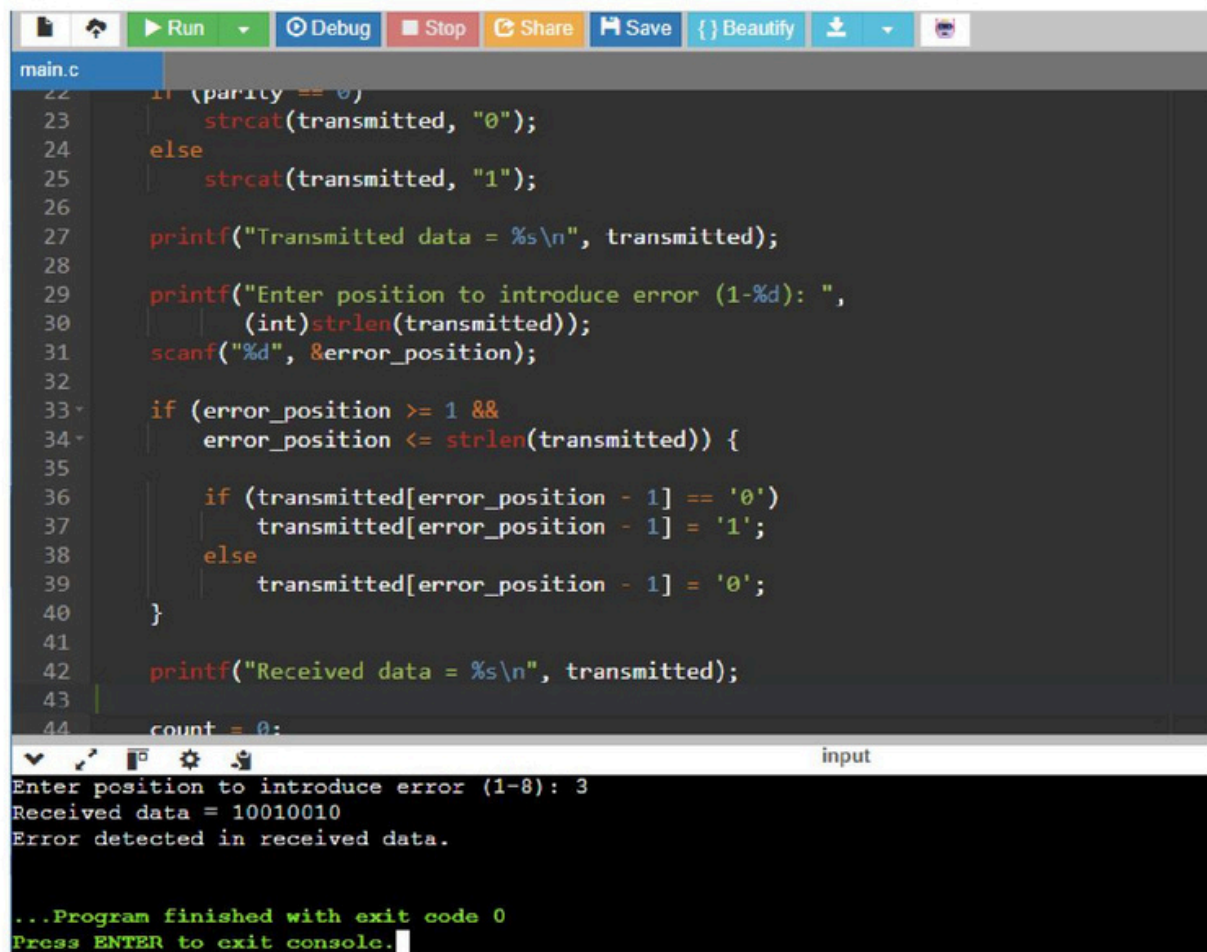
Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:
 0  164 ms  3 ms  168 ms  172.25.96.1
 1  15 ms  212 ms  19 ms  172.22.10.1
 2  24 ms  49 ms  6 ms  14.139.187.225
 3  *  *  *  Request timed out.
 4  *  *  *  Request timed out.
 5  *  *  *  Request timed out.
 6  *  *  *  Request timed out.
 7  *  *  *  Request timed out.
 8  86 ms  9 ms  20 ms  72.14.213.20
 9  48 ms  100 ms  26 ms  216.239.43.133
10  113 ms  77 ms  113 ms  142.251.56.63
11  256 ms  26 ms  25 ms  dns.google [8.8.8.8]

Trace complete.

C:\Users\monica>

```

QUESTION 4:



The image shows a code editor window with a file named `main.c`. The code is a C program that checks for parity and introduces an error at a specific position. The code is as follows:

```
22 if (parity == 0)
23     strcat(transmitted, "0");
24 else
25     strcat(transmitted, "1");
26
27 printf("Transmitted data = %s\n", transmitted);
28
29 printf("Enter position to introduce error (1-%d): ",
30       (int)strlen(transmitted));
31 scanf("%d", &error_position);
32
33 if (error_position >= 1 &&
34     error_position <= strlen(transmitted)) {
35     if (transmitted[error_position - 1] == '0')
36         transmitted[error_position - 1] = '1';
37     else
38         transmitted[error_position - 1] = '0';
39 }
40
41
42 printf("Received data = %s\n", transmitted);
43
44 count = 0;
```

Below the code editor is a terminal window. The output of the program is as follows:

```
Enter position to introduce error (1-8): 3
Received data = 10010010
Error detected in received data.

...Program finished with exit code 0
Press ENTER to exit console.
```